1 Summary

Fuselage Section 13 — Forward — Door Surround Structure

Pre-Analysis — Titanium Corner Fitting Standard at the Fwd Cargo Door Surround

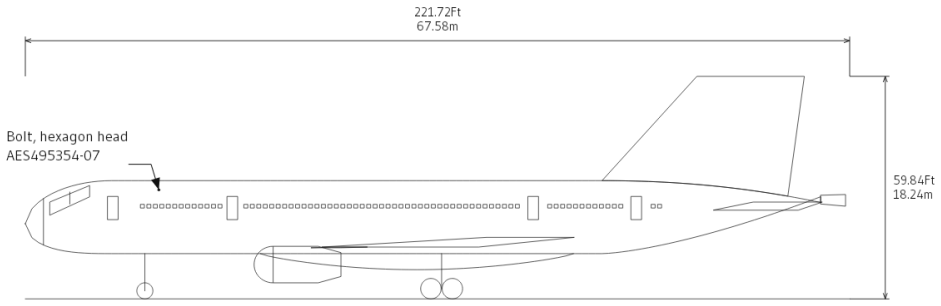
Object code	DMC-MRD-A-53-21-13-00A-019A-A	**Issue**	001-00 (new)
Issue date	2018-04-19	**Language**	en-GB
Responsible partner	AERALIS S.A.S. (U8025)	**Originator**	AERALIS S.A.S. — Pre-Analysis, Neuhafen (U8025)
Security	01 — Unclassified	**Quality assurance**	First verification (tabtop)
Applicability	MERIDIAN-42 and MERIDIAN-46, all aircraft		

1 Summary

Closed pre-analysis for the titanium corner fitting standard at the forward cargo door surround. Four items were confirmed, one of them blocking on the in-service inspection bulletin whose threshold was written against the aluminium fitting.

2 Objective

Establish, ahead of design freeze, what the change from the aluminium corner fitting to the titanium alloy standard invalidates in the existing in-service package. The design office asked specifically whether the recurring inspection already mandated at the lower corner survives the change, whether the illustrated parts content can carry two fitting standards side by side, and whether the crack stop hole repair scheme written for the frame lower web remains usable once the fitting footprint moves inboard by eleven millimetres.



Fuselage Section 13 — Forward — Door Surround Structure — location, side view — Figure 1

3 Findings summary

Four items were confirmed. The blocking one is the recurring detailed inspection: its threshold and repeat interval were derived from the aluminium fitting crack growth curve, and a post-modification aircraft inspected on that interval would be inspected far more often than the substantiation requires, at real cost to the operator and with no safety benefit. The

4 Next steps

illustrated parts content and the removal task both carry a single fitting item and a single fastener release sequence, so both need a pre and post pair rather than a revision in place. The frame lower web repair scheme was the surprise: its stop hole positions are dimensioned from the fitting flange edge, and the new flange edge sits inboard, so two of the three permitted hole positions fall outside the permitted land.

4 Next steps

Raise a revision request against the inspection bulletin so the post-modification threshold is issued at the same time as the modification, not afterwards. Ask the repair design owner to re-dimension the stop hole positions from a datum that does not move with the fitting. Agree the two-standard item numbering with technical publications before the parts content freeze, and confirm with in-service engineering that no aircraft is part way through the inspection campaign at cut-in.

5 Work order data

Table 1 Work order data

Work order	PAWO-160142-01
Subject modification	MOD-160142
Status	Completed
Outcome	Validated
Raised	2017-07-03
Due	2018-03-30
Completed	2018-04-19
Analysis round	1
Lead department	ESZS
Lead site	Neuhafen
Contributing departments	EDSA, EYIS
Man-hours estimated	130
Man-hours spent	168
Recommendation	Proceed with conditions

Table 2 Analysis tasks

Task	Title	Method	Owner	Status
T01	Retrieve every released item on ATA 53-21-13 and the door surround interface	Entity and semantic	ESZS	Done
T02	Check the recurring inspection threshold against the new fitting substantiation	Manual review	EYIS	Done
T03	Assess the frame lower web repair scheme against the revised flange edge	Entity	EDSA	Done

6 Applicability and effectivity

Table 3 Findings

Ref	Document	Type	Severity	Rationale
F01	MRD-53-1008	Effectivity invalidation	Blocking	The one-off and repeat detailed inspection in this bulletin is threshold-driven from the aluminium fitting crack growth curve. Post-modification aircraft carry a fitting with a different stiffness and a different residual stress state, so the interval is no longer representative and the bulletin says nothing about the new standard.
F02	SRM-M-53-21-11-002	Repair conflict	Major	The crack stop hole positions in the frame lower web repair are dimensioned from the corner fitting flange edge. The titanium fitting moves that edge inboard, so two of the three permitted positions fall outside the permitted land and the repair cannot be applied as written to a post-modification aircraft.
F03	IPC-M-53-21-13-FIG-01	Manual impact	Minor	The figure carries one corner fitting item and one fastener set. Two standards will be in service simultaneously for the whole retrofit period, so the figure needs a pre and post item pair and a matching effectivity note.
F04	AMM-M-53-21-13-000-001-A	Manual impact	Minor	The removal task was reached by following the impact edge from the earlier fitting standard rather than by text similarity. It carries a single fastener release sequence and a torque table that does not cover the titanium fastener set.

Closed. The inspection bulletin threshold was the pacing item and was re-issued in step with the modification; the repair scheme was re-dimensioned. The change went to release with the documentation package complete.

6 Applicability and effectivity

MERIDIAN-42 and MERIDIAN-46, all aircraft

7 Referenced documents

Table 4 Referenced documents

Document	Type	Relationship
MOD-160142	MOD	Pre analysis of